

Paradigmatic, not dyadic:

Revisiting the role of semantic transparency in complex word processing

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Background and objectives Semantic transparency – the extent to which the meaning of a morphologically complex word can be inferred from its structure and constituents – has been extensively investigated as a key factor in the representation and processing of derived words [1, 2, 3, 4, 5, a.o.]. However, most studies have approached this issue from a dyadic perspective, focusing primarily on the relationship between a derived word and its lexical base. More recent theoretical developments in derivational morphology have instead advocated a paradigmatic approach, according to which derived words should be analyzed not only in relation to their bases but also with respect to the morphological paradigms in which they are embedded [6]. The present study investigates the role of semantic transparency from this paradigmatic perspective and compares its effects with those observed under the traditional base-derivative approach.

Method We examine the semantic transparency of derived words using *Démonette-2.0*, a large-scale derivational lexicon of French [7], and distributional measures based on `word2vec` embeddings trained on the FR-COW16 web corpus. We consider three complementary measures: (i) base-derivative relatedness (T_{base}), the cosine similarity between the embeddings of a derivative and its base; (ii) derivational family relatedness (T_{fam}), the mean cosine similarity between the embeddings of a derivative and all words sharing its stem; and (iii) process typicality (T_{proc}), the mean cosine similarity between the embeddings of a derivative and all words formed through the same morphological process. To evaluate the psycholinguistic relevance of the three measures, we assess their ability to predict lexical decision data from the French Lexicon Project (FLP; [8]) using a sample of $n = 3,982$ lemmas with complete data. We compare three generalized additive mixed models with random intercepts for process and family: a baseline model including lemma frequency, length, ambiguity, attestation date, family size, series size, and morphological productivity as control predictors (Model 1); a model extending the baseline with T_{proc} and T_{base} (Model 2); and a model extending the baseline with T_{proc} and T_{fam} (Model 3).

Results Models 2 and 3 significantly improve on Model 1 in predicting both standardized reaction times (zRT) and error rates (ER), confirming that transparency, broadly construed, contributes to the prediction of lexical decision performance (see Figure 1). The effect of T_{proc} is linear, robust, and highly significant in both models and for both outcomes ($p < .001$), and holds even among derivatives produced by the same process. T_{fam} has a strong nonlinear effect on both outcomes ($p < .001$), emerging at high levels of family relatedness. Crucially, Model 3 fits the data substantially better than Model 2 ($\Delta\text{AIC} = -22$ for zRT; $\Delta\text{AIC} = -31$ for ER), despite the two models being otherwise identical in structure and covariates.

Discussion The results indicate that the paradigmatic dimensions of transparency – process typicality and derivational family relatedness – have a significant impact on lexical processing, supporting a paradigmatic account of transparency and, more broadly, of morphological organization in the mental lexicon. Furthermore, the results suggest that semantic transparency in derivation is not a unitary construct, but rather comprises distinct dimensions that differently affect morphological processing. Interestingly, semantic transparency appears to exert an inhibitory effect in non-priming lexical decision tasks, contrasting with the facilitative effect reported in priming lexical decision experiments. This contrast underscores the importance of task demands in evaluating the psycholinguistic consequences of linguistic factors [9].

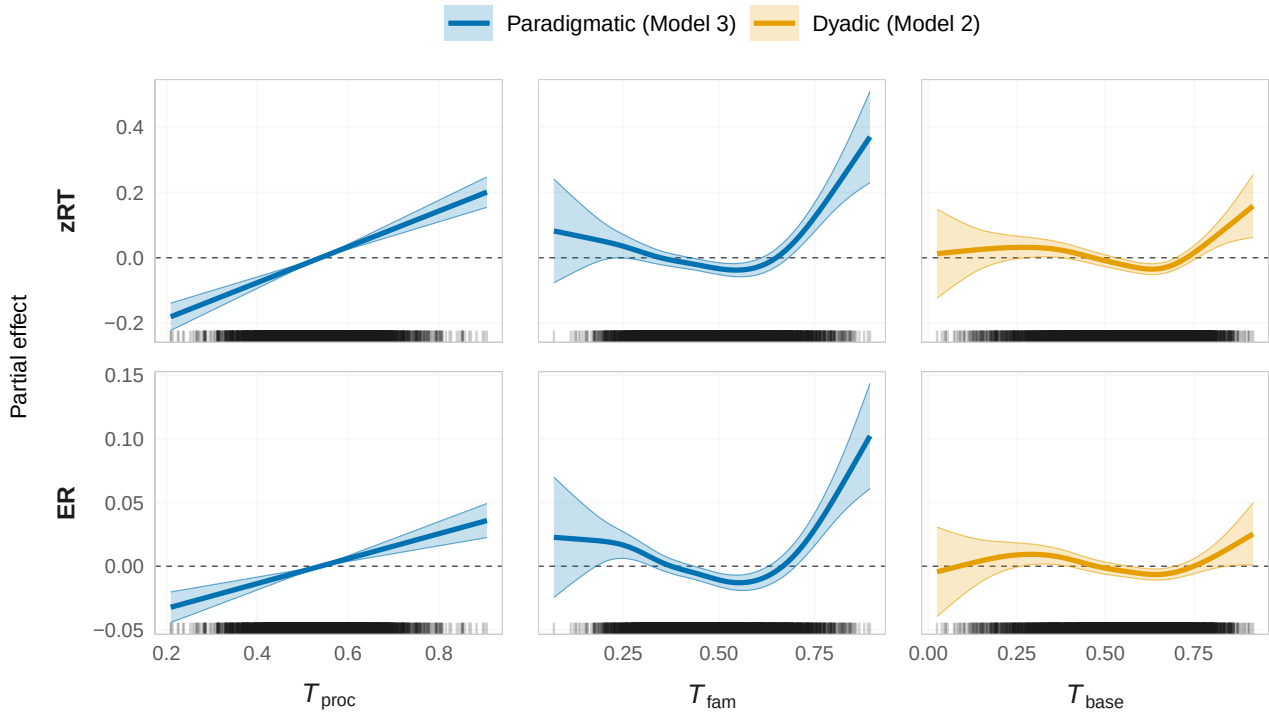


Figure 1: Partial effects (95% CI) of T_{proc} and T_{fam} (Model 3, paradigmatic) and of T_{base} (Model 2, dyadic) on standardized response times (zRT) and error rates (ER), estimated by generalized additive mixed models.

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